

// PORTFOLIO · 2026

Febri Rizki

UI/UX Designer & AI Engineer

Combining intuitive user interface design expertise with production-grade artificial intelligence engineering. B.Sc. & M.Sc. background in Biology (USU 2024 & Unsyiah 2018) provides strategic advantage for AI solutions in life sciences & healthcare domains. Focused on building impactful end-to-end digital products — from user research to AI model deployment in production.

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// README.md

v1.0.0 · 6 repositories

About Me

I am a UI/UX Designer and AI Engineer with an academic background in Biology (M.Sc. from Universitas Sumatera Utara, 2024 & B.Sc. from Universitas Syiah Kuala, 2018) that proves to be a strategic advantage in developing AI solutions — particularly for domains requiring understanding of complex systems, data analysis, and analytical thinking patterns. I combine user research, interface design, and AI model integration to deliver experiences that are both intuitive and backed by reliable technology. Currently focused on building generative AI-powered platforms, MLOps, and intelligent mobile applications. My approach is always user-centric — every technical decision should serve user needs, not the other way around.

// Core Focus

- UI/UX Design & Design System
- AI Engineering & LLM Integration
- MLOps & Automated Retraining
- Frontend Engineering (Next.js, React)
- Mobile ML (Flutter + TensorFlow Lite)

// Education

- M.Sc. Biology - USU (2024)
- B.Sc. Biology - Unsyiah (2018)
- Computational Biology
- Bioinformatics & Data Science
- Scientific Method & Statistics

Tech Stack

Below are the technologies I actively use in building digital products across UI/UX, frontend, AI/ML, and deployment & DevOps. Every project in this portfolio uses a combination of the technologies listed below.

TypeScript

JavaScript

Python

Dart

React

Next.js 16

Tailwind CSS 4

Flutter

TensorFlow Lite

Google Gemini

MLflow

GitHub Actions

PostgreSQL

Prisma ORM

Convex

Docker

Vercel

Cloudflare Pages

Featured Projects

Below are 10 featured projects showcasing the combination of UI/UX Design and AI Engineering expertise — from AI-powered design platforms to automated MLOps pipelines. Each entry is presented like a GitHub repository page with tech stack badges, code snippets, and key metrics. All repository links are accessible for verification.



AI-Powered Design Platform with BranchBoard

A collaborative design platform that integrates generative AI into the UI/UX workflow. BranchBoard allows designers to create parallel design variations with AI-assisted creative exploration — similar to Git branching, but for visual assets.

● TypeScript ● Next.js 16 ● React 19 ● Tailwind 4 ● Prisma ● Neon Postgres
● Gemini API ● dnd-kit ● @mdxeditor

- ▶ **BranchBoard:** visual branching system for parallel design exploration with AI-generated variations
- ▶ Drag-and-drop design canvas with dnd-kit for reusable components
- ▶ MDX editor for integrated inline design documentation
- ▶ Serverless Neon Postgres + Prisma for global scale without overhead
- ▶ Generative AI assistant for component & layout alternative suggestions

12+

UI COMPONENTS

AI

POWERED

0.2.0

VERSION

[→ github.com/hyperchainpro/hyperchainpro](https://github.com/hyperchainpro/hyperchainpro)

Adaptive AI Tutor for High-School Students

A personalized adaptive AI tutor for high-school students, built for the **Build with Gemini XPRIZE 2026** hackathon. Uses a Socratic approach to guide students through course material with auto-generated quizzes, smart flashcards, essay review, and a math solver.

● Next.js 16 ● TypeScript ● Prisma ORM ● PostgreSQL ● Gemini 2.0 Flash
● Tailwind 4 ● shadcn/ui ● Framer Motion ● Zustand

- ▶ **Adaptive Learning Profile:** onboarding captures grade, target exam, subjects, and daily goal
- ▶ **AI Tutor Chat:** Gemini-powered chat with Socratic approach that adapts to student understanding level
- ▶ **Smart Flashcards:** spaced repetition with SM-2 algorithm for long-term retention
- ▶ **Essay Review:** AI scoring with rubric-based feedback for essay writing
- ▶ **Math Solver:** step-by-step worked solutions for math problems
- ▶ **12 Achievements:** gamification system to keep learning motivation high

```
// Auto-generated quiz by Gemini AI per subject + topic
const response = await gemini.chat.completions.create({
  model: 'gemini-2.0-flash',
  messages: [{ role: 'system', content: socraticPrompt }],
});
```

12

ACHIEVEMENTS

6+

AI FEATURES

XPRIZE

HACKATHON

[→ github.com/febririzki95/pathmentor-ai](https://github.com/febririzki95/pathmentor-ai)



AI-Powered Cooking Assistant

An AI-powered cooking assistant that helps users discover recipes, calculate nutrition, and get cooking suggestions based on available ingredients. Integrates Convex for real-time backend and Cloudflare Pages for global edge deployment.

● Next.js 16

● TypeScript

● Convex

● Prisma ORM

● Gemini AI

● shadcn/ui

● Tailwind 4

● Cloudflare Pages

● Hook Form

- ▶ **Recipe Generation:** AI generates recipes based on ingredients available in the user's kitchen
- ▶ **Nutrition Tracking:** Gemini AI integration for automatic nutrition analysis
- ▶ **Real-time Sync:** Convex backend for real-time recipe updates across devices
- ▶ **Edge Deployment:** Cloudflare Pages with OpenNext for global performance
- ▶ **Smart Forms:** Hook Form + Zod for valid & structured recipe input

```
// Edge deployment via OpenNext → Cloudflare Pages
scripts: {
  build: 'next build',
  build:cloudflare: 'opennextjs-cloudflare build',
  pages:deploy: 'wrangler pages deploy .open-next/assets'
}
```

Real

TIME SYNC

Edge

GLOBAL DEPLOY

0.2.0

VERSION

→ github.com/hyperchainpro/dapurmindai



Automated ContentRewards Campaign Pipeline

An end-to-end automation system for ContentRewards: campaign scraper, automated video editor, and a social poster to 34 platforms simultaneously. Designed for content creators who want to maximize content distribution without manual intervention.

● Next.js 16

● TypeScript

● Prisma ORM

● Tailwind 4

● shadcn/ui

● dnd-kit

● @mdxeditor

● Playwright MCP

- ▶ **Campaign Scraper:** automatically fetches latest campaign data from ContentRewards
- ▶ **Video Editor:** automated template-based video editing with MDX editor for scripts
- ▶ **34-Platform Social Poster:** distributes content to 34 platforms at once (YouTube, TikTok, Instagram, etc.)
- ▶ **Playwright MCP:** browser automation for posting to platforms without official APIs
- ▶ **Standalone Build:** Next.js standalone output for lightweight Docker deployment

```
// Standalone build + custom server for long-running workers
build: 'next build && cp -r .next/static .next/standalone/.next/',
start: 'NODE_ENV=production bun .next/standalone/server.js'
```

34

PLATFORMS

Auto

PIPELINE

0.2.1

VERSION

[→ github.com/hyperchainpro/cr-autopilot](https://github.com/hyperchainpro/cr-autopilot)



Flutter ML App · 2023 Food Categories

A Flutter app that uses Machine Learning (TensorFlow Lite) to identify food types from photos. Can recognize 2,023 food categories with inference running on a background thread (Isolate) for optimal performance, integrated with MealDB API & Gemini AI for nutrition information.

Flutter

Dart

TensorFlow Lite

Gemini AI

MealDB API

image_cropper

GitHub Actions

- ▶ **2023 Food Categories:** TFLite model can recognize 2,023 food types with high accuracy
- ▶ **Background Thread Inference:** TFLite runs in Isolate to keep UI responsive
- ▶ **Multi-source Intelligence:** MealDB for recipes & ingredients, Gemini AI for nutrition & descriptions
- ▶ **Smart Image Picker:** camera + gallery + image_cropper for optimal results
- ▶ **Clean Architecture:** modular separation of models/services/screens/widgets/utils

```
// Background thread ML inference
Future<FoodPrediction> predict(File image) async {
  return await compute(_runInference, image.path);
  // TFLite via Isolate - no UI freeze
}
```

2023

CATEGORIES

224²

INPUT SIZE

2

EXT APIS

[→ github.com/febririzki95/food_recognizer_ci_check](https://github.com/febririzki95/food_recognizer_ci_check)



Automated ML Model Retraining with MLflow + GitHub Actions

An MLOps pipeline that automates the machine learning model retraining process using MLflow Project and GitHub Actions CI. Uses the Breast Cancer Wisconsin dataset as a proof-of-concept, with Docker Hub deployment for environment reproducibility.

● Python ● MLflow ● GitHub Actions ● Docker ● Conda ● scikit-learn

- ▶ **MLflow Project Spec:** standardized MLproject file for model training reproducibility
- ▶ **CI/CD Pipeline:** GitHub Actions workflow automatically runs `mlflow run MLProject --env-manager=local`
- ▶ **Environment Reproducibility:** `conda.yaml` + Docker image for dependency isolation
- ▶ **Modular Scripts:** separate `modelling.py` for training, `automate_febri.py` for preprocessing
- ▶ **Docker Hub Integration:** image published for deployment to other environments

```
# .github/workflows/workflow_ci.yml
name: ML Retraining CI
on:
  schedule: [{ cron: '0 0 * * 0' }] # weekly
  workflow_dispatch:
jobs:
  retrain:
    runs-on: ubuntu-latest
    steps:
      - run: mlflow run MLProject --env-manager=local
```

Auto

RETRAINING

CI/CD

PIPELINE

Docker

REPRODUCIBLE

[→ github.com/febririzki95/Workflow-CI](https://github.com/febririzki95/Workflow-CI)

AI Platform with Caddy Server & Neon DB

A production-grade AI platform using Caddy Server as reverse proxy, Neon DB serverless PostgreSQL, and Prisma ORM. Supports dual database (PostgreSQL & SQLite) for deployment flexibility, with MDX editor for inline documentation and dnd-kit for drag-and-drop interactions.

● Next.js ● TypeScript ● Prisma ORM ● Neon DB ● Caddy Server ● dnd-kit

● MDX Editor ● SQLite

- ▶ **Production Architecture:** Caddy reverse proxy + auto HTTPS for enterprise deployment
- ▶ **Dual Database Support:** PostgreSQL (Neon serverless) & SQLite for local/dev
- ▶ **Mini-services Architecture:** modular service separation in `mini-services/` folder
- ▶ **MDX Documentation:** inline editor for integrated technical documentation
- ▶ **Test Suite:** dedicated `tests/` folder for regression testing

2x

DATABASE

Caddy

PROXY

0.2.1

VERSION

[→ github.com/febririzki95/NeuroPilot](https://github.com/febririzki95/NeuroPilot)



ML Preprocessing Automation · Breast Cancer Dataset

Supervised Machine Learning experiment repository focused on preprocessing the Breast Cancer Wisconsin dataset. Covers Exploratory Data Analysis (EDA), data cleaning, feature engineering, and reproducible automation scripts for research-grade results.

● Python ● Jupyter Notebook ● scikit-learn ● Pandas ● GitHub Actions
● NumPy

- ▶ **EDA Notebook:** complete exploratory statistical analysis with feature distribution visualization
- ▶ **Automated Preprocessing:** Python script for reproducible data transformation
- ▶ **Breast Cancer Wisconsin Dataset:** benchmark dataset from UCI ML Repository
- ▶ **CI Automation:** GitHub Actions workflow for automated preprocessing validation
- ▶ **Cleaned Output Dataset:** CSV ready for model training in Workflow-CI

```
# Automated preprocessing pipeline
import pandas as pd
from sklearn.preprocessing import StandardScaler

df = pd.read_csv('breastcancer_raw/data.csv')
df_clean = automate_febri.preprocess(df)
df_clean.to_csv('breastcancer_preprocessed.csv', index=False)
```

EDA

NOTEBOOK

Auto

PREPROCESS

UCI

DATASET

→ github.com/febririzki95/Eksperimen_SML_Febri



Integrated Digital Ecosystem Landing Page

Landing page for the HyperChain Pro digital ecosystem featuring 3D animations with Three.js, motion design via GSAP & Framer Motion, and edge deployment through Cloudflare Pages. Demonstrates UI/UX capabilities with interactive visualizations and immersive product storytelling.

● Next.js 16 ● React 19 ● TypeScript ● Tailwind 4 ● Three.js ● GSAP
● Framer Motion ● Neon DB ● Prisma 7 ● Cloudflare Pages

- ▶ **3D Visualizations:** Three.js for interactive hero section and product showcase
- ▶ **Scroll Animations:** GSAP ScrollTrigger for cinematic product storytelling
- ▶ **Component Transitions:** Framer Motion for smooth micro-interactions
- ▶ **Edge Deployment:** Cloudflare Pages with pnpm for efficient builds
- ▶ **Database Integration:** Neon Postgres + Prisma 7 for lead capture & newsletter

3D

VISUALS

Edge

DEPLOY

10+

ANIMATIONS

→ github.com/hyperchainpro/hyperchain-landing



Multi-platform Document Converter App

A multi-platform document converter app built with Expo (React Native) supporting conversion of PDF, DOCX, XLSX, and images to various formats. Uses Supabase for backend, expo-camera for document scanning, and rich libraries like jsPDF, docx, exceljs, and mammoth for file parsing & generation.

Expo

React Native

TypeScript

Supabase

jsPDF

docx

exceljs

mammoth

Cloudflare Pages

- ▶ **Multi-format Support:** convert between PDF, DOCX, XLSX, JPG, PNG, and many more
- ▶ **Document Scanner:** expo-camera + image manipulation for physical document scanning
- ▶ **Biometric Auth:** expo-local-authentication for sensitive data security
- ▶ **Supabase Backend:** real-time file storage sync & user management
- ▶ **Web Export:** build to web via expo export --platform web for cross-platform access
- ▶ **Sharing Integration:** expo-sharing to share files directly to other apps

```
// expo export for cross-platform
scripts: {
  build:web:prod: 'expo export --platform web --output-dir=dist',
  start: 'expo start',
  android: 'expo start --android',
  ios: 'expo start --ios'
}
```

Multi

FORMAT

3x

PLATFORM

1.0.0

VERSION

[→ github.com/hyperchainpro/hyp-convert](https://github.com/hyperchainpro/hyp-convert)

Awards & Contact

★ Semifinalist, Hackathon PIDI Digdaya BI 2026

Bank Indonesia · 2026 · National

Collaborated with a team to develop an innovative data- and AI-driven digital solution supporting Bank Indonesia's digital transformation agenda. Passed the preliminary selection round and ranked among the top 20 teams nationally out of hundreds of participating teams.

// CERTIFICATIONS

Certifications

Machine Learning & AI

4 certificates

Dicoding Academy

Frontend & UI/UX

5 certificates

Dicoding Academy

Backend & Cloud

5 certificates

Dicoding Academy

Mobile Development

4 certificates

Dicoding Academy

Big Data & Data Analytics Training (BDT)

1 professional certificate

BDT Program

// CONTACT

Let's Connect

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\$ certs	drive.google.com/...

// Thank you for reviewing my portfolio.
// Let's create something impactful together.